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CROWD DETECTION IN A BUS

ABSTRACT

The public transportation system, especially buses, has been noted to suffer from problems associated with controlling the passenger load, hence making it hard for the commuter to enjoy a safe ride. This mini project proposes a crowd detection system for buses using the YOLO object detection technique together with a camera module. The system collects video feeds through a camera placed inside the bus and processes individual frames through the YOLO object detection model to detect the number of passengers in each frame and determine the crowd density level (low, medium, or high). The information regarding the count and crowd status is presented on a dashboard or a central server, which transport managers and commuters use to monitor the system. The benefits of this system include better planning of routes and informing passengers of the crowd status at specific bus stops. YOLO was selected because it is known to be both fast and accurate in object detection, which makes it ideal for real-time applications. The proposed solution is expected to be affordable as it uses a regular camera along with a lightweight YOLO object detector such as YOLOv5 or YOLOv8 nano.

Keywords: YOLO; Object Detection; Crowd Detection; Real-Time Monitoring; Passenger Counting; Computer Vision; Embedded Systems; Raspberry Pi; Public Transportation; Smart Transportation System

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